LESSON PLAN

EHM (28)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	7	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Sheet metal repair - patch repair, stop drilling of cracks.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Sheet metal repair - patch repair, stop drilling of cracks.
5. Specific Objectives	28.1.3.4 Trainee will be able to explain Sheet metal repair procedure.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of patch repair preparation and stop drilling technique; Safety precautions regarding PPE and tool handling; Instructor supervision of drill alignment and hole spacing.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary causes of sheet metal fatigue cracks?
(b) List three safety precautions required before starting sheet metal repair.
(c) Define the term 'stop drilling' in the context of crack repair.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Assessment of crack extent and location • Preparation of surface for patch application • Selection of patch material and fasteners • Drilling of stop holes at crack tips • Application of patch and fastening sequence	L & Ds	PPT, IB	15 min
(b)	• Execution of stop drilling procedure • Placement and sizing of patch material • Fastening sequence and torque application • Inspection of repair integrity • Final surface finishing and sealing	L & Ds	PPT, PDF	20 min
(c)	• Common defects in patch repairs • Troubleshooting fastener loosening issues • Documentation of repair records • Review of maintenance manual procedures • Safety review of repair zone	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the critical importance of stop drilling in preventing crack propagation.
(b) Describe the correct sequence for applying a patch repair to ensure structural integrity.
(c) Identify two potential failure modes if the patch is not fastened according to the specified torque.

12. Home Assignment and Follow Up	Prepare a brief report outlining the safety precautions and step-by-step procedure for composite repair as covered in the next lesson plan (LP 07), highlighting key differences from sheet metal repair.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Composite repair - procedure and safety precautions while working of composites.	Refer to the master syllabus reference list	LP 07

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min